

Sustainable accessibility: a conceptual framework to integrate transport and land use plan-making. Two test-applications in the Netherlands and a reflection on the way forward

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Abstract

The integration of transport and land use planning is widely recognized as essential to the achievement of sustainable development. The concept of accessibility—or what and how can be reached from a given point in space—can provide a useful conceptual framework for this integration. More specifically, a shift of focus in urban transport planning from catering for mobility to catering for accessibility helps see how more sustainable transport options can, under certain land use conditions, provide a competitive degree of accessibility that matches less sustainable options. The authors have used the concept of accessibility as a framework for the interactive design of integrated transport and land use plans in two areas of the Netherlands. The objective of these exercises was identifying solutions where economic, social, and environmental goals could be combined, defined as the achievement of ‘sustainable accessibility’. The existing situation has been evaluated, and alternative plans have been developed. In this paper we reflect on these experiences and sketch the way forward, with a focus on the methodological aspects of the undertaking. In this respect, a major challenge is finding a workable balance between an accessibility measure that is theoretically and empirically sound and one that is sufficiently plain to be usefully employed in interactive, creative plan-making processes.

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1. Introduction

The integration of transport and land use planning is widely recognized as an essential, but often neglected, precondition of sustainable development (e.g. Wegener and Fürst, 1999; Priemus et al., 2001; Meyer and Miller, 2001). The concept of accessibility—or what and how can be reached from a given point in space—can provide a useful framework for this integration. If adequately defined, accessibility can be directly related to both the qualities of the transport system (e.g. travel speed) and the qualities of the land use system (e.g. functional densities and mixes). At the same time, it can be directly related to economic goals (access to workers, customers, suppliers), social goals

(access to employment, goods and services, social contacts) and environmental goals (resource-efficiency of the associated activity and mobility patterns). In particular, a shift of focus in urban transport planning from catering for mobility to catering for accessibility may help see how more sustainable travel options (e.g. walking, cycling, public transport, shorter car trips) can, under certain land use conditions (e.g. higher densities, more finely-tuned functional mix), provide a degree of accessibility that matches less sustainable options (Bertolini and le Clercq, 2003).

In this paper, we discuss two test-applications of this approach in the Netherlands. In the first example accessibility measures have been used to support the design of integrated transport and land use plans for the Rotterdam-The Hague region; in the second example as a way of gaining insight in the interdependency between transport and land use options in the Delta Metropolis (a more recent term for the Randstad, the highly urbanized West of the country). The objective of the first exercise was identifying

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transport and land use solutions where economic, social, and environmental goals could be combined, defined as the achievement of ‘sustainable accessibility’. In the second exercise transport and land use plans under discussions have been scrutinized in order to identify both the inconsistencies and potential synergy between transport and land use interventions. In both cases the objective was not so much an evaluation of the effects of policy measures (for this purpose there are many available tools), but rather assisting the joint design of transport and land use plans (a far less developed field). In the last part of the paper we reflect on these experiences and explore the way forward, with a focus on the methodological aspects of the undertaking. In this respect, the major challenge has proven to be that of finding the right balance between an accessibility measure that is theoretically and empirically sound and one that it is sufficiently plain to be usefully employed in interactive, multi-actor plan-making processes.

2. Research and policy background

The two projects under discussion are related to the broader research programme ‘Transportation performance for the region’ (in Dutch ‘VervoersPrestatie voor de Regio’, or VPR)¹. Aim of VPR is developing a method for integrating transport and land use plan making at the urban-regional scale within the overall goal of enabling more sustainable mobility. Before describing the two applications, some assumptions and definitions behind VPR will be introduced.

2.1. The VPR process: constructing development strategies and programmes

As far as the VPR process is concerned two points are essential. The first is the characterization of policy-making as a network of interconnected, but not necessarily sequential actions involving distinct stakeholders. The second is the recognition of a lack of tools for supporting the design of policies, as opposed to the relative abundance of tools for analysing the problems at hand and evaluating alternative solutions. The scheme in Fig. 1 can be employed to clarify both these points. The policy-making process is structured by the interrelated actions of three distinct groups of actors: politicians, civil servants and professionals, and their respective constituencies. As far as politicians are concerned, and moving from the more abstract to the more concrete, key actions entail identifying and matching relevant issues, acceptable interventions, concrete projects

and measures and available budgets. Professionals rather think in terms of visions, concepts, plans and projects. At least in the Dutch context, it is especially civil servants that are responsible for linking the political and the professional: they provide an input to the whole process by characterizing situations in terms of trends and attitudes towards them; they are responsible for translating problems and visions in policy goals; interventions and concepts in development strategies; measures and plans in development programmes; and budgets and projects in actual implementation.

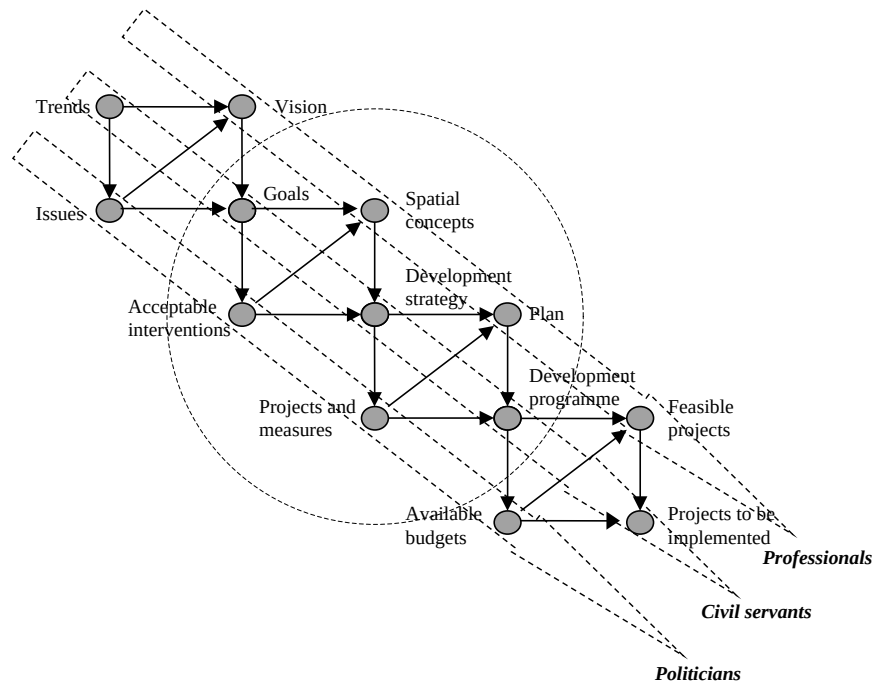
Actual policy making rarely moves along the process represented in Fig. 1 in a linear, top-down fashion. Rather, at any given moment actions are underway at all levels and in all domains. Recognizing the interrelatedness of such actions, and thus the need for improving consistency between them can be seen as an essential condition for the success of policies. Tools for supporting such effort in the upper and lower levels of the process are relatively well developed (think at different sorts of surveys and demand and supply analysis techniques on one side and at plan evaluation and project prioritisation methods on the other side). However, tools for the interactive design of ‘strategies’ and ‘programmes’ are much less developed. These are the main focus of VPR.

2.2. The VPR goal: shaping conditions for sustainable accessibility

The overall aim of VPR is improving the sustainability of urban regional mobility. However, in most real-world situations sustainability will, certainly if defined in pure environmental terms, never be the only and often not even the dominating policy goal, but will rather be weighed against other goals, such as economic competitiveness or social equity. This means that solutions that succeed in serving more goals at the same time will invariably be the ones with the greatest chance of success. In particular, the combination of the goals of improving sustainability and accessibility appears central to overcoming the current friction among major environmental issues, social aspirations and economic imperatives (see for instance WBCSD, 2001). The alternative solution is defining sustainability in broad terms, that is, jointly accounting for environmental, social, and economic aspects of sustainability. However, we contend that for analytical purposes it is still useful to distinguish between an ‘environmental’ sustainability and an ‘economic’ and ‘social’ accessibility. But what should we precisely understand under ‘sustainability’ and ‘accessibility’?

Direct measures such as energy use, CO₂ emissions, air pollution, traffic noise and the like would be the best indicators of the environmental impact of urban transport. However, these measures are rarely available, making the per-capita distance travelled by car a widely accepted (un)sustainability indicator of urban transport (Wegener and Fürst, 1999). This indicator is—at least in the present

¹ We gratefully acknowledge the Ministry of Economic Affairs and the Ministry of Transport, Public Works and Water Management of the Netherlands for financing through the Netherlands Agency for Energy and the Environment (NOVEM) our contribution to this program. For more documentation on VPR see Meurs (2004).



and near future technological context—highly correlated with the above-mentioned negative environmental impacts. From this perspective, the challenge for sustainable urban transport and land use planning becomes that of shaping transport and land use conditions where the potential for the use of other, environmentally friendlier transportation methods than the conventional car is maximized.

In order to adequately define accessibility the introduction of three broadly supported assumptions about human behaviour is required (see among others Hägerstrand, 1970; Zahavi, 1974; Downes and Emmerson, 1985; Schafer and Victor, 1997; Wiel, 2002):

- People travel for the most part not just for the sake of it, but in order to participate in spatially disjointed activities (e.g. living, working, shopping, or visiting in different places)
- People want to have a choice among as large a number and as diverse a range of activities as possible
- Travel costs, and—at least in the developed world—most importantly travel time, rather than travel distance, set a limit to these possibilities (in the form of total daily travel time budgets, travel-to-work time budgets etc.)

Following these assumptions accessibility can be defined as ‘the amount and the diversity of places of activity that can be reached within a given travel time and/or cost’, and the overall VPR goal of maximizing the synergy between sustainability and accessibility as the goal of:

Developing transport and land use conditions for as large as possible a share of environmentally friendlier transportation

methods than the conventional car, while at the same time maintaining and possibly increasing the amount and the diversity of activity places that people can reach within a given travel time and/or cost.

Environmentally friendlier transportation (or better ‘activity coupling’) methods might include not travelling (access provided by information technology), walking, cycling, public transport and the use of more efficient (cleaner) cars. Some of the transport and land use policy implications are sketched in Fig. 2. A more thorough discussion is in Bertolini and le Clercq (2003).

The assumptions and definitions introduced above will sound hardly revolutionary to many. Furthermore, there can be discussion on their appropriateness. However, they have proved a very useful, even needed starting point for a plan-making process aiming at acknowledging and building upon the interdependency between transport and land use interventions, and involving participants who are rather used to consider the two aspects separately. The two examples discussed below will give more substance to this claim. But before doing that, let us first introduce the accessibility measure adopted.

3. Selecting an appropriate accessibility measure

Conceptualisations of accessibility along the lines discussed in the preceding section have long informed the academic and planning debate. However, the translation of such concepts in performance measures that can be usefully employed to improve integration of transport and land use

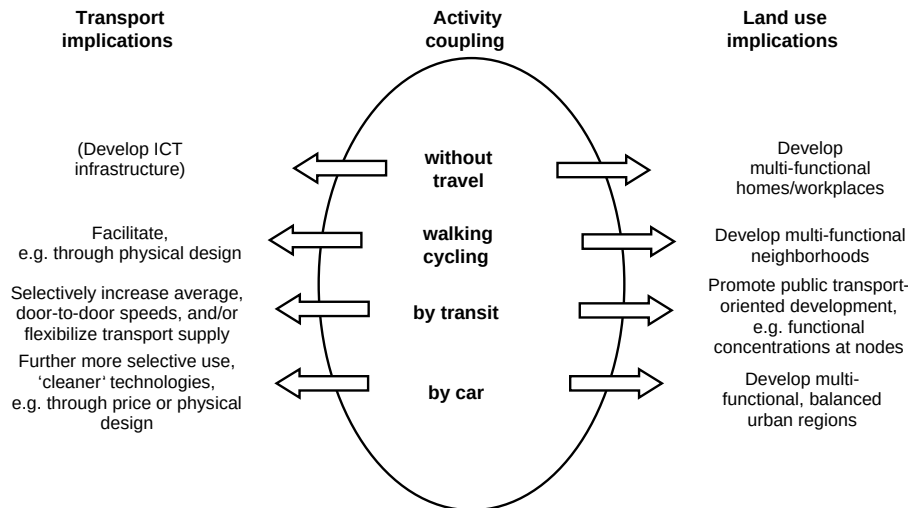


Fig. 2. Policy implications of accessible and sustainable urban form.

plan making in practice is still very limited (Handy and Niemeier, 1997; Geurs and van Wee, 2004). In order to be useful for practical planning purposes an accessibility measure must meet two basic requirements: it must be consistent with the uses and perceptions of the residents, workers, and visitors of an area, and it must be understandable to those taking part in the plan-making process. There are great differences in the way available accessibility measures strike a balance between these two requirements. Accessibility measures with a transport and land use component can be divided in three broad types: cumulative opportunities measures, gravity-based measures, and utility-based measures (Handy and Niemeier, 1997; Geurs and Ritsema van Eck, 2001; Halden, 2002; Geurs and van Wee, 2004). Roughly, they are characterized by both ascending behavioural consistency and difficulty of interpretation. In the applications discussed below, we opted for the simplest type: a cumulative opportunities measure, expressed in terms of spatial opportunities to be reached within a given travel time. There are a number of specification, calibration, and interpretation issues that need to be addressed when selecting an accessibility measure (Handy and Niemeier, 1997). In the following, our choice will be discussed in relation to each of these issues.

Issues of specification include the degree and type of disaggregation, the definition of origins and destinations, and the measurement of attractiveness and travel impedance. The basic unit of analysis in our studies was the neighbourhood as defined by the Central Bureau of Statistics (CBS). In the Netherlands, this is a rather fine-grained spatial unit to which a wealth of readily available land use, socio-demographic and economic data is attached. The analysis focussed on home to work trips and, more implicitly, on access to non-daily services. Home to work trips are, in the Netherlands as elsewhere, a minority and declining share in total mobility. However, because of their highly constrained nature they

have a more than proportional impact on the day-to-day functioning of an urban region (see Wiel, 2002 for a similar contention based on the exceptional frequency and duration of the activity 'working'). The concentration of non-daily services (shopping, leisure, public facilities) has historically characterized the urban core. In emerging multi-centred urban regions non-daily services are, however, being redistributed from the old city centre to a number of new centres, making access to them a key area of concern. Spatial opportunities were defined two perspectives: that of the competitive position of firms in terms of potential employees and customers within reach, and that of the welfare position of households in terms of access to the main concentrations of employment and non-daily services. Only home-based, single purpose, unimodal trips were considered. While there is legitimate concern about the appropriateness of such choices in the literature (Handy and Niemeier, 1997; see also Wilson, 2000; and Meyer and Miller, 2001), many trips still fall within this category, at least when access to work and to non-daily services is considered. In the Netherlands, only train use shows a significant degree of intermodal trip chaining (other than walking to a transit stop or a parking place), with 33% of railway passengers biking to the station, 30% walking, 22% using another public transport means, 10% the car, and 5% other modes (Fietsverkeer, 2003). The quality of access to and from railway stations by walking, cycling and other public transport means has accordingly been an issue, albeit qualitative, in the applications. Travel impedance has been measured in terms of travel time. For this purpose maximum allowed speeds on the different sections of the road network have been coded, as have allowances for speed reduction on congested roads. Similarly, average speeds have been assigned to all sections of the railway network (with a distinction between intra- and interregional services) and to the main metro, tram and regional bus lines. Speeds for the remaining public transport

and for walking and cycling have been averaged on the basis of straight-line distances. Such separate consideration of travel modes was essential to the objective of assessing the competitive position of alternatives under different transport and land use conditions.

The key calibration issue when adopting a cumulative opportunities measure is the choice of the cut-off travel distance or time. The literature does not, however, provide a clear rule on how to make this choice (Handy and Niemeier, 1997). In this study we adopted a 30-min travel time limit. There were two main reasons for setting the time limit at 30 min. The first one is empirical. In the Netherlands, 80% of all commuters spend half an hour or less on their home to work trip. The average commuting time is 28 min (in 2001, according to the Central Bureau of Statistics). This is an average in line with international figures (see for instance Kenworthy and Laube, 1999). Furthermore, there appears to be a meaningful link between this figure and the actual functioning of urban regions, at least in the context of middle-sized European cities. Prud'homme and Lee (1999) document a robust relationship between the labour productivity of French middle-sized cities and an 'effective labour market' defined in terms of a 25-min home to work travel time limit. Wiel (2002) observes that the functional limit of French urban regions, measured in terms of impact on real estate values, is at about half of the daily travel time budget from the main concentrations of employment opportunities, or around 30 min. The second reason for choosing a fixed travel time limit stems from the overall purpose of the study and may at least partly offset the inevitable arbitrariness of a hard cut-off travel time relative to a behaviourally more consistent continuous impedance function. The overall purpose of the study was helping identify transport and land use conditions favourable to sustainable accessibility. In this respect, the 30 min limit should not be so much seen as an absolute norm, but rather as a benchmark against which to assess the potential impact of changes in transport and land use features. Are the latter, all else being equal, shaping conditions for shorter or longer travel times? And thus, do they reinforce or weaken the position of an urban centre or the attractiveness of a residential location (in terms of access to, respectively, potential employees and customers or jobs and non-daily services)? In the same vein, how do they modify the competitive position of alternative modes of travel?

The final set of issues addresses matters of interpretation. This was, because of our ambition of supporting an interactive, creative policy design process a paramount area of concern. Following Forester (1989), we see design as 'making sense together through practical conversations', and aim at developing analytical tools 'structuring' such process. This was perhaps the main reason for choosing a cumulative opportunities measure, which is easy to understand and nevertheless already allows dealing with essential

trade-offs and interdependencies between interventions in the transport and in the land use system. For the same reason, car and public transport accessibility were analysed separately, as was the impact of road congestion, accessibility by cycling and walking, and the spatial distribution of changes in accessibility. As documented in the next two sections, the choice proved in this respect to be an adequate one. Of course, the above does not overcome the fundamental consistency limits of a cumulative opportunities measure. Further development of the approach requires dealing with these limits more adequately. Some possible solutions are discussed in Section 6. But let us first turn to the two applications. In both cases, attention should be directed not only, perhaps not so much, to the results, but also, perhaps rather, to the process through which they have been achieved.

4. Designing sustainable accessibility in the Rotterdam-The Hague area

The first application (le Clercq et al., 2003) is a pilot project within the VPR programme in the Rotterdam-The Hague area. The exercise had no formal policy goal but rather aimed at experimenting with the new method and increasing awareness of the potential of an integrated land use and transport plan-making approach. More specifically, the scope of the pilot was fourfold: first, testing the new method in a (quasi) real-world, interactive plan-making process; second, looking for synergies between sustainability objectives and current national transport and land use policies; third, exploring ways of linking national policy objectives to transport and land use choices at the urban regional level; fourth, making the possibilities of the VPR approach better known to national government officials. Officials of the Ministry of Transport, Public works and Water management and of the Ministry of Housing, Spatial Planning and the Environment were involved in the pilot, as well as representatives of the VPR sponsors.

Relative to the scheme in Fig. 1, the pilot focussed on the identification of a 'development strategy', and on ways of interactively linking the 'professional' and 'civil servant' lines of actions. More specifically, it consisted of the following phases:

- A study of policy documents and a round of interviews to identify current national and regional goals and approaches in the fields of urban development and mobility;
- A kick-off meeting to compare and link national policy objectives, the general aim and approach of VPR, and specific issues in the urban region under study;
- A first design workshop to interactively develop alternative transport and land use options for the area;
- A second design workshop to evaluate, discuss and further define the two options;

- A third design workshop to interactively define a preferred development strategy combining the strengths of the two alternatives;
- A presentation of the results to a broader public, most notably including local stakeholders (mainly regional and municipal authorities), to discuss possible implications.

The interactive process began with the specification of the goal of the plan-making exercise. This, while inspired by the general assumptions and definitions discussed in Section 2 of this paper, was amended following interviews with national government officials and discussions with participants in the workshops. Such interactive goal-definition has proved essential: the ability to connect the pursuit of sustainable accessibility as defined within the VPR approach to the thrust of existing national and regional policies showed a crucial factor in granting proactive involvement of national government officials in the project. More specifically, the enhancement of accessibility could be connected to current national policy goals of ‘economic functionality’ (granting firms access to potential employees and customers), ‘freedom of choice’ (providing citizens with access to a variety of jobs and services) and ‘equity’ (ensuring that such access is evenly distributed). All of these goals are aptly captured by the definition of accessibility introduced in previous sections. Such a definition proved also useful when discussing potential policy measures, because it allows a direct link to both transport and land use measures. Accordingly, and perhaps most importantly, it provides a way of shedding light on the interdependency and the trade-offs between these measures, as it shows how accessibility can be improved by either transport improvements, land use improvements or a combination of the two.

Also valuable were discussions on the definition of sustainability. In particular, it proved essential that more aspects of sustainability were included. This was obtained by defining sustainable accessibility as accessibility ‘with as little as possible use of non renewable, or difficult to renew, resources, including land and infrastructure’. Most notably, the characterization of land and infrastructure as scarce resources—only implicit in the more general formulation discussed in the preceding section—allowed a direct link to current national government concerns, and this at a time when ‘narrow’ sustainability (i.e. in environmental terms only) is quite low on the agenda. Also the definition of sustainability provided a useful link to potential policy measures, as it was possible to show its connection with the use of the most resource-efficient modes of transport and with the average distance travelled.

After there was agreement on the definition of sustainable accessibility, we could move on to address the task of finding ways of improving it. Starting point was the analysis of accessibility in the present situation, with a focus on the position of the main concentrations of workplaces

and non-daily services—or, in Dutch policy jargon, ‘urban nodes’—relative to residential areas. This was done by first drawing lines connecting all the points to be reached within a 30 min travel time—or isochrones—from these urban nodes for various transport modes and conditions (car, car with congestion, public transport), and by then computing inhabitants that fell within these contours. Figs. 3 and 4 show, as an example, the isochrones for the urban nodes of Rotterdam Central Station and The Hague Central Station. Tables summarizing the main results of this analysis for all urban nodes in the region—similar to Table 1 below—were also produced, as well as ‘reverse pictures’—similar to Fig. 7 below—showing the amount of urban nodes that can be reached from a given residential area.

The results of the analysis, and especially discussing them with workshop participants, paved the way to the solution-searching process: how could the accessibility of both urban nodes and residential areas be improved in a sustainable way? A first agreement was reached that—as far as mobility at the urban regional scale was concerned—improving the competitive position of public transport should be a central part of the effort. In accessibility terms this meant looking for ways of expanding the public transport isochrones of the main urban nodes and/or increasing land use densities and functional diversity within them. However, three conditions to this search were also set: first, the need to take account of already existing or planned public transport infrastructure, rather than assuming new one will be developed in the short term; second, the need to acknowledge that most people are not dependent on public transport, and will also have to, or just want to, be able to use the car; third, the need to acknowledge the multiple scales of accessibility, and to therefore consider the implications of choices for mobility at higher and especially lower scales than the urban-regional. Also these three conditions could be translated in the language of isochrones: the first implies giving priority to intensification and diversification of uses within existing public transport isochrones as opposed to expanding such isochrones; the second requires looking at the patterns of accessibility by not only public transport but also by car; the third points to the need of considering implications of transport and land use interventions for the isochrones of bicycle and walking (for instance: does a proposed railway link sever local bike routes and thus ‘compress’ bicycle isochrones?).

With these and analogous assumptions in mind and tools in hand, two groups went on to design each a set of transport and land use policy measures in the form of an integrated transport and land use plan for the region. While there was awareness of the importance of ‘soft’ measures such as parking policy or road pricing, it was decided to focus in this phase on ‘hard’ measures such as the location of urban development relative to infrastructure. Intriguingly enough, and as typical in design processes, the same premises led to different solutions. One group gave priority to the consolidation of activity and mobility patterns in

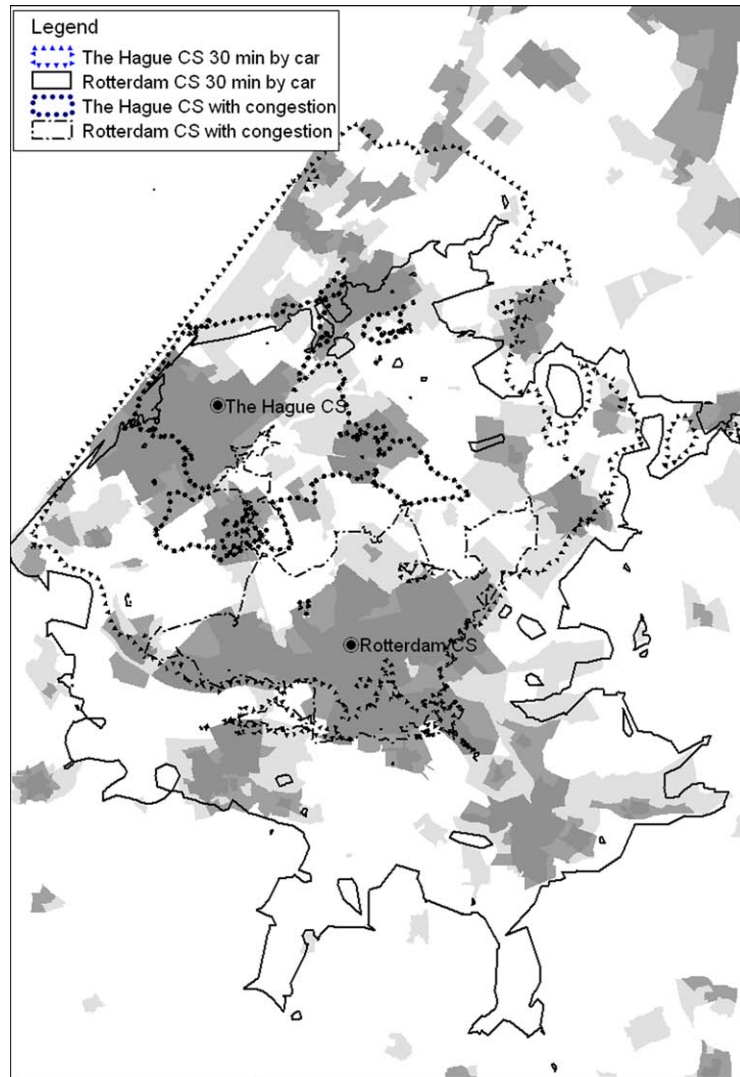


Fig. 3. Isochrones of The Hague Central Station and Rotterdam Central Station by car, free-flow and with congestion.

and around the existing main agglomerations of Rotterdam and The Hague, the other group to the development of public transport corridors between the two. There were, however, also many points of contact between the proposals, including the deliberate choice for a number of 'car environments' in areas where improving public transport accessibility and/or increasing land use densities was deemed to be unrealistic. The accessibility change in both plans was then analysed with the same approach adopted for the existing situation. In both cases the need to improve (public) transport links was evident. However, the first plan showed that, at least potentially, significant enhancement of accessibility could also be reached by land use measures only (i.e. concentration and diversification of functions). This is important, as recent experience in the Netherlands documents how little certainty there is that planned infrastructure be realized, and if so, on time.

Following a discussion of the analysis of the two initial transport and land use models a preferred option was

developed, building on the strong points of both: consolidation in and around existing centres where most feasible and development of the public transport corridors with the highest (expected) benefits to costs ratios. This final model is represented in Fig. 5.

What are the most important conclusions of this experience? A first point is that taking part in the design process greatly enhanced the participants' awareness that accessibility—while positively related to a number of current policy goals—can be improved by either transport policies (improving average speeds) or land use policies (increasing density and fine-tuning the functional mix). In the end, there was agreement that the challenge is that of developing an optimal combination of the two, improving both accessibility and sustainability while taking into account context constraints (such as available financial resources and political conditions) and local specificities (such as existing infrastructure and urban morphologies and landscape structures). As far as the methodology



Fig. 4. Isochrones of The Hague Central Station and Rotterdam Central Station by public transport.

is concerned, participants greatly appreciated the use of isochrones for their ability of making the interdependency and trade-offs between transport and land use measures visible and debatable. At the same time however, there was also recognition of the need of more sophisticated tools to evaluate the actual impacts of the alternatives, and thus move on to further stages in the policy-making process.

5. Discussing accessibility in plans for the Delta Metropolis

Following participation in the Rotterdam-The Hague pilot the Transport Research Centre of the Ministry of Transport, Public works and Water management (MoT) asked the authors to apply the approach to an analysis of land use options for the Delta Metropolis (also known as the Randstad, the heavily urbanized west of the country, including the four major cities of Amsterdam, Rotterdam,

The Hague and Utrecht, see Fig. 6). This pilot (Bertolini et al., 2003) was indirectly, rather than directly connected to actions in the scheme of Fig. 1. The MoT was involved in discussions with the Ministry of Housing, Spatial Planning and the Environment on a preferred 'development strategy' for the Delta Metropolis and was looking for ways of making mobility implications of land use choices more visible. Central to the debate was the question of where to locate new residential areas, and the implication of this choice for (1) the hoped for strengthening of the competitive position of existing concentrations of jobs and facilities (or urban nodes) and (2) the just as fervently desired efficient use of the existing transport infrastructure.

The approach was roughly the same as that followed in the Rotterdam-The Hague VPR pilot: drawing isochrones from all the main urban nodes, computing residents and jobs within these isochrones, and analysing the implications of changes in transport provision (or the shape of isochrones) and land use development (or the amount of residents

Table 1
Inhabitants and jobs within reach of urban nodes in the Delta Metropolis, existing situation

Urban nodes	Inhabitants $\times 1000$, within 30 min			Jobs/workers, within 30 min	
	By car, free flow	By car, congestion	By public transport	By car, congestion	By public transport
<i>South wing</i>					
Delft	2396	335	663	1.27	1.29
Den Haag CS	2446	628	800	1.24	1.14
Den Haag HS	2341	579	867	1.21	1.14
Den Haag Mariahoeve	2057	249	569	1.81	1.23
Dordrecht	1589	316	355	0.94	1.18
Gouda	2117	264	409	1.01	1.17
Leiden	1581	329	646	0.86	1.25
Rotterdam Alexander	2850	785	562	1.08	1.15
Rotterdam CS	2520	809	878	1.11	1.09
Rotterdam Lombardijen	2240	784	638	1.15	1.11
Schiedam	2779	681	835	1.22	1.18
Voorburg	2771	494	559	1.46	1.25
Zoetermeer	2820	300	555	1.55	1.22
<i>North wing</i>					
Amsterdam Amstel	2757	813	648	1.40	1.47
Amsterdam CS	2094	590	814	1.28	1.30
Amsterdam Lelylaan	2101	589	846	1.29	1.26
Amsterdam Sloterdijk	2814	991	594	1.44	1.69
Amsterdam Zuid	2279	776	819	1.45	1.37
Amsterdam Bijlmer	2910	789	716	1.52	1.58
Haarlem	1894	396	422	1.06	1.15
Hoofddorp	2486	507	271	1.44	2.08
Schiphol	2717	665	538	1.45	1.68
Zaandam	1917	325	426	0.94	1.20
<i>East wing</i>					
Almere Centrum	1127	132	186	0.69	0.96
Amersfoort	1495	251	246	1.06	1.20
Hilversum	1697	211	343	1.03	1.28
Maarssen	1963				
Utrecht CS	1757	363	534	1.53	1.40
Utrecht Leidsche Rijn	2368	487	164	1.37	1.96
Utrecht Uithof	1809	424	223	1.47	1.63

The distinction in three zones—or wings—is made out of convenience, and has no implications for the analysis (i.e. also accessible jobs and inhabitants beyond the borders of each zone are computed).

and jobs within a given isochrone). First, the existing situation was analysed. Table 1 summarizes the results as far as urban nodes are concerned. There were three main conclusions. First, there is a great variation in the amount of inhabitants and jobs within reach from different urban nodes. These differences in accessibility from urban nodes point at differences in the size of their spatial markets (potential employees, customers, business relations within reach) and thus in their development potential. A critical point in this respect is that urban nodes on the edge of the main cities tend to show a better position than areas around central railway stations (CS) of the major cities. Second, Table 1 shows differences in the vulnerability to congestion of urban nodes and in the competitive position of public transport. The policy implication suggested is that—from an accessibility point of view—development of workplaces and facilities at urban nodes that score badly or one of both of these points should not be encouraged, as it will likely lead to a relatively less efficient use of the existing road infrastructure and to a disproportionate growth of (longer) car trips. Alternatively,

enough should be done to manage congestion and improve public transport provision at these urban nodes. The opposite applies to urban nodes with higher relative scores on these two aspects: reinforcing the central function of these locations will be likely to contribute to manage congestion and enhance the potential use of public transport. Third, there are differences in the proportion of jobs to working population within the isochrones of the urban nodes. Some nodes have a relative surplus of working population within their potential labour market area, others a deficit, with differences of up to 40%. The first situation suggests opportunities to develop workplaces and facilities within the isochrone, or alternatively the need to restrain residential developments there. The second situation suggests the opposite, or the need to restrain growth of jobs and services, unless more houses are built within the isochrones.

The maps in Fig. 7 can help summarize the results of the analysis as far as new residential locations were concerned. The first observation concerns the position of new residential areas relative to urban nodes. When accessibility

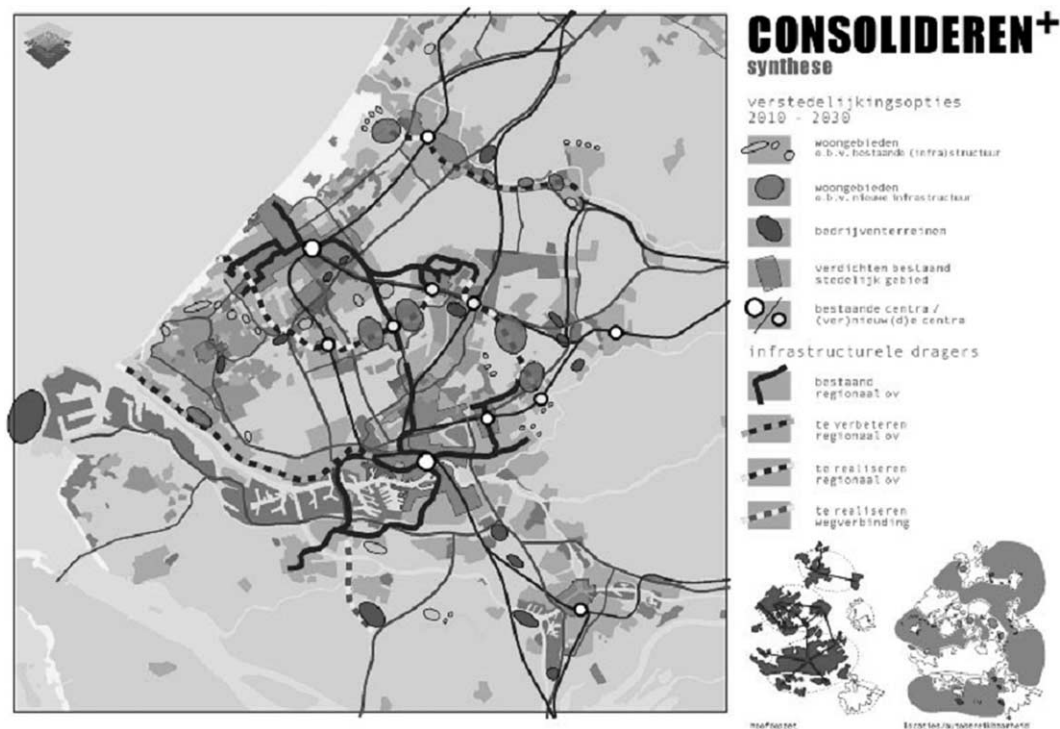


Fig. 5. The preferred model, a combination of concentration of residential and other urban functions in and around the two major existing conurbations and development of selected public transport corridors (or rather 'necklaces') between them. Existing built-up area, light grey; New urban developments, dark grey; Transport and urban development corridors, thick lines and dark grey ovals.

by car is considered it is areas along the motorway corridors between the largest cities that score best. If congestion is accounted for it is rather areas at the periphery of those same cities that appear to be well placed. On the other hand, areas

directly adjoining the central urban nodes score relatively better if accessibility by public transport is considered. The latter is, however, much worse in absolute terms (i.e. less urban nodes can be reached within the same travel

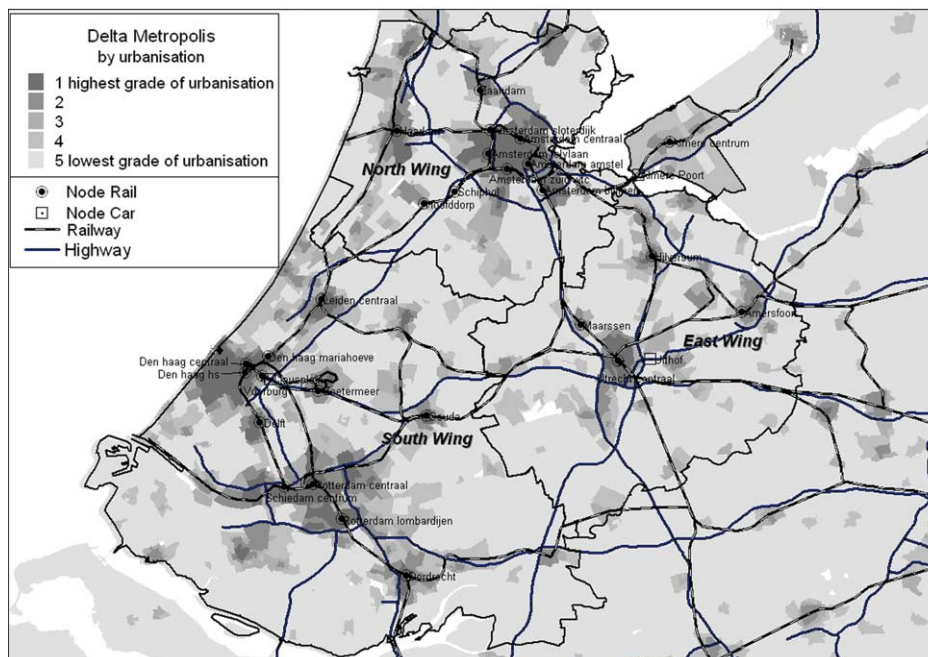


Fig. 6. Urban nodes in the Delta Metropolis, as identified by current national and regional policy documents.

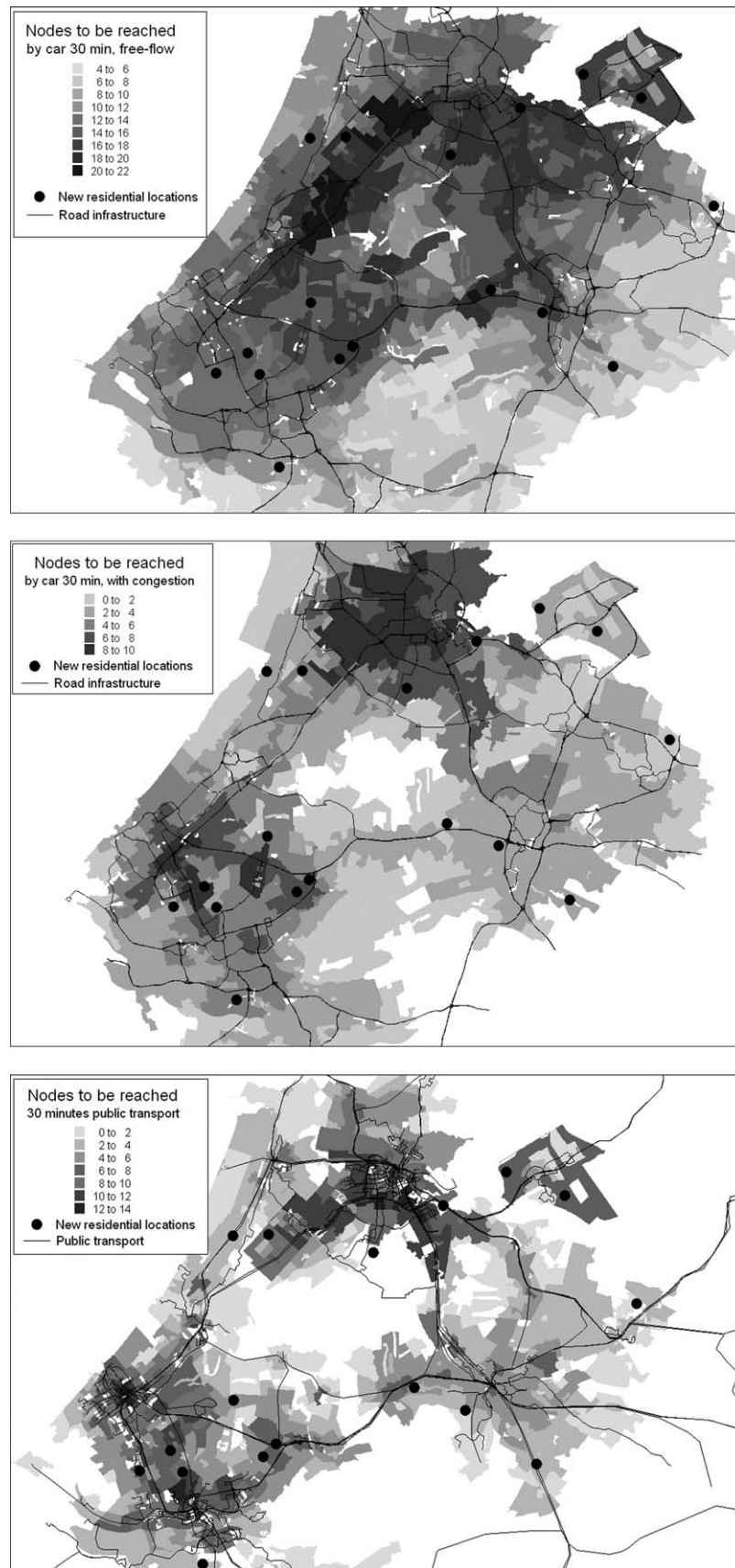


Fig. 7. Urban nodes within 30 min travel time by different transport means and conditions, after infrastructure improvements. From top to bottom: by car free-flow, by car with congestion, by public transport. Dots represent proposed residential locations.

time). It can be expected that less accessible residential areas will not be very desirable, or that people choosing to live there—because of reasons other than accessibility—will have to travel longer. At the same time, such residential areas will be likely to contribute less to strengthening the position of the urban nodes, because they will be beyond their ‘natural’ reach. The opposite applies for more accessible residential areas.

A second observation concerns the contribution of new residential areas to improving the workers-jobs balance, the idea being that it is desirable to enhance chances for matching jobs and workers at shorter travel times and distances. The most significant conclusion—and next to implications for individual locations—is that residential areas in the job-rich, dynamic northern and eastern parts of the Delta Metropolis are, in accessibility terms, much more needed than residential areas in the economically more stagnant south (around Rotterdam and The Hague). This is a fact that raises interesting questions about future development paths: will new residential areas in the south be less desirable because of their lower job accessibility or will rather long commuting patterns towards others part of the Delta Metropolis emerge? And is there a possibility (as some contend but many dispute) that jobs will follow people and move to the south? The answer depends, of course, on a number of factors. But also from a policy point of view there are, depending on the objectives, more than one preferred outcome. In other words, these are questions that the analysis above cannot answer. What the analysis does show, is rather the existence of interdependencies and trade-offs between land use and transport developments that need to be accounted for: for instance by acknowledging the unlikelihood of certain combinations, such as short commutes without a significant shift of homes toward the north and east and/or of jobs towards the south. Bringing this sort of awareness in the plan-making process is precisely the objective of the approach introduced here. This last point is an apt introduction to the methodological discussion in Section 6.

6. On measuring accessibility: soundness or plainness?

A major methodological challenge when working with accessibility measures in land use and transport planning is finding the right balance between a measure that is theoretically and empirically sound and one that is sufficiently plain to be usefully employed in interactive, creative plan-making processes where participants typically have different degrees and types of expertises. As far as soundness is concerned consistency of the measure with the behaviour of households and firms is the central concern. As far as plainness is concerned transparency of computing procedures and ease of calculations—or the communicative qualities of the measure—are essential. This second requirement is not just because of a generic democratic

concern, but also because of the importance of mobilizing the (tacit) knowledge of different participants in the identification of problems and the search for solutions. In the choices discussed above plainness was the paramount preoccupation. However, a price was paid in terms of soundness. In the following, directions for striking a better balance are explored, with implicit reference to the ongoing debate about the appropriate accessibility measures for strategic planning purposes (e.g. Handy and Niemeier, 1997; Geurs and Ritsema van Eck, 2001; Halden, 2002; Geurs and van Wee, 2004).

In the Rotterdam-The Hague and in the Delta Metropolis study a simple cumulative opportunities measure was used (the amount of inhabitants, jobs or urban nodes to be reached from a location within a given travel time). As far as soundness is concerned there are two main problems with this measure. First, there is the question of how to define the cut-off travel time. A contour sets a sharp borderline between areas within reach and those outside, whereas in reality the transition will be much more fluid, as people might accept longer travel times when this allows them to access very attractive opportunities or shorter in the opposite case. Furthermore, degrees of acceptance will differ depending on individual features and trip purposes. In our study we have applied a 30-min time limit based on national data on home to work trips. A more accurate time limit can be based on travel budget studies, which consider time spent for specific travel purposes as a proportion of total time spent on specific out-of-home activities (e.g. Dijst and Vidakovic, 2000; Schwanen and Dijst, 2001; Wiel, 2002). It can also be based on utility functions, which can be estimated as part of a destination choice model (Breheny, 1978; see also Wilson, 2000). In the latter case, the parameter accounting for the generalized cost component can be interpreted as the willingness to incur certain travel costs. This willingness can be translated in acceptable travel times per trip purpose, or travel times that are accepted by a certain percent of a particular travelling population.

The second major problem with the measure adopted is that it does not deal with competition among spatial opportunities. This is a crucial point if the functioning of urban spatial markets is to be adequately captured. For instance, the position of a retail outlet will not only be determined by the amount of potential customers within an acceptable travel time, but also by the degree of competition from other retail outlets for those same customers. In another example, the position in the labour market of workers will not only be influenced by the amount of jobs at an acceptable travel time from their homes, but also by the degree of competition from other workers for those same jobs. It is in this respect important to be precise about what is meant by competition. It might be the competition for scarce resources, as in the second example. A way to incorporate this form of competition in accessibility measures is—as for instance suggested by

Geurs and Ritsema van Eck (2001)—the application of the balancing factors of the well-known single and double constrained spatial interaction models. Another form of competition is that related to the freedom people have to choose to travel to a different destination, depending on the attractiveness of the location and travel costs to get there. This amounts to competition among locations for the people making their destination choice, as in the example of retail outlets. A gravity-based measure can be applied in this second case, but the area of influence should be reduced following the likely number of travellers going to competing destinations. This reduction can be treated similarly to the way road congestion is treated in transport models or the way comparisons are made between accessibility by car and by public transport. In general, it will correspond to the probability other locations will be chosen, as can be derived from a destination choice model when the attractiveness of all destinations would be similar.

In summary, an improved way to apply accessibility measures in a plan-making context where plainness is important, but theoretical and empirical insights must not get lost could be the following, in increasing order of complexity:

1. Use travel time, or better generalized travel costs contours that are sensitive to the travel purpose and to individual socio-demographic characteristics. Acceptable travel costs or times can be derived from travel costs or time utility estimates with a logit model. Such a contour measure could be best employed for quick-scan or learning purposes, such as gaining insight in the interdependencies between land use and transport patterns and policy measures. Using a cost rather than a time contour could furthermore provide a useful way of visualizing the effect of ‘soft’ measures, such as road pricing or parking fees. The main advantages are that the measure is easily explainable, and does not require assumptions about a person’s perception of transport, land use and the interaction of these two; and that the necessary data are readily available, including the possibility to select in the analysis particular kinds of people or types of activity.
2. This accessibility measure can be further improved by considering, instead of the sharp limits of a contour (e.g. more or less than 30 min), a more gradual decrease in travel time or cost utility, that is, adopting a gravity-based accessibility measure.
3. As far as taking account of competition effects is concerned, a distinction between sorts of competition is important. In the case of competition for limited resources at destinations, origins need to be redistributed among destinations taking account of constraints at those destinations. In the case of competition for limited resources at origins, reduction in the areas of influences

identified in 2 could be applied, representing the probabilities that other destinations are chosen.

7. Conclusions: the way forward

The goal of this paper was showing how the concept of sustainable accessibility could be employed for the integration of transport and land use planning in an interactive plan-making process. Two applications have been discussed: the first aiming at developing an integrated transport and land use plan for enhancing sustainable accessibility in the Rotterdam-The Hague area, the second providing information for structuring the discussion around the transport implications of land use options for the Delta Metropolis. In both studies a simple cumulative opportunities measure was used. The use of this measure has proved an apt tool for enhancing insight in the interdependencies and trade-offs between land use and transport patterns and policy measures in the course of an interactive plan-making process. A cumulative opportunities measure is able to capture implications for both economic, social and environmental policy goals of given land use and transport patterns and show how land use and transport measures can be combined in order to achieve these goals. Ease of understanding is without doubt one of the greatest strengths of this measure. However, its theoretical and empirical weaknesses should be also recognized. In future applications contours based on acceptable travel times or costs derived from utility functions can help overcome the arbitrariness involved in setting a generic time limit. Furthermore, differentiation according to socio-demographic characteristics and trip purposes could be treated in this way. In the second place, in order to take account of the effects of competition on accessibility, reduction in areas of influence representing the probability that other destinations are chosen and/or capacity constraints at destinations can be applied.

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